

LESSON 6-8 • SESSION 2 • PRACTICE • FORM A

Expanding and Factoring Expressions

Learning target – I can expand an expression and factor one using the greatest common factor. · 6.EE.A.3

REMEMBER

- To **expand**, multiply the outside number by each term inside: $3(2x + 5) = 6x + 15$.
- To **factor**, pull out a number every term shares. Factoring **undoes** expanding.
- Use the **GCF**: $8x + 20 = 2(4x + 10)$ is not finished, because 4 and 10 still share 2. $8x + 20 = 4(2x + 5)$ is.

WORKED EXAMPLEFactor $18x + 27$ using the GCF.**Step 1** The GCF of 18 and 27 is 9.**Step 2** Divide each term: $18x \div 9 = 2x$ and $27 \div 9 = 3$.**Step 3** Expand to check: $9 \times 2x + 9 \times 3 = 18x + 27$.**Answer** $9(2x + 3)$ **WORDS** **expand** (*desarrollar*) · **factor an expression** (*factorizar una expresión*) · **greatest common factor** (*máximo común divisor*)**PRACTICE**

show your work · label units

- 1.**
- Expand each expression.

a. $8(c + 3) =$ _____

b. $5(2d + 4) =$ _____

c. $6(y + 7) =$ _____

d. $7(n + 2) =$ _____

- 2.**
- Factor each expression using the GCF.

a. $12m + 20 =$ _____

b. $15k + 35 =$ _____

c. $14b + 35 =$ _____

d. $18x + 24 =$ _____

- 3. Ticket Booths.**
- Booth 1 writes its ticket cost as
- $4(t + 9)$
- . Booth 2 writes its cost as
- $20t + 45$
- .

Hint Find the GCF of 20 and 45 first.

a. Expand Booth 1's cost. _____

b. GCF of $20t$ and 45 : _____

c. Factor Booth 2's cost. _____

Show your work.

- 4. Expand, Then Combine.**
- Simplify
- $3(x + 10) + 2x$
- .

Hint 3 times 10 is part of the answer.

Show your work.

Simplest form _____

- 5. Expand, Then Combine.**
- Simplify
- $4(z + 6) + 5z$
- .

Show your work.

Simplest form _____

6. **Nora’s Factoring.** Nora factored $16y + 40$ as $2(8y + 20)$. Is she done? If not, factor with the GCF.

Hint Do 8 and 20 still share a factor?

Show your work.

GCF form _____

7. **Both Directions.** Factor $30c + 42$ using the GCF. Then expand your answer to show the two forms match.

Hint The GCF of 30 and 42 is 6.

Show your work.

Factored form _____

8. **Garden Rows.** A garden has $24p + 36$ plants.

- a. Factor the expression using the GCF. _____
- b. What could the factored form tell you about rows? _____
- c. Find the number of plants when $p = 4$. Use both forms. _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Lucia’s Work.** Lucia expanded $5(w + 8)$ and wrote $5w + 8$. She said the 8 was already inside the parentheses. Explain her mistake. Then expand correctly.

TEST PRACTICE

choose the best answer

10. Which is $10x + 15$ factored using the GCF?
A $5(2x + 15)$ **B** $5(2x + 3)$ **C** $10(x + 5)$ **D** $2(5x + 15)$
11. Which is the expanded form of $6(y + 4)$?
A $6y + 4$ **B** $6y + 24$ **C** $10y$ **D** $y + 24$

SELF-CHECK CODE

find your answer · copy its letter into that problem’s box · spell a math word

132	5x + 30	5x + 10	9z + 6	5	10d + 20	8(2y + 5)	9z + 24	4(3m + 5)
S	T	Q	Z	C	F	R	O	A

#1b	#2a	#3b	#4	#5	#6	#8c

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____